

stage1_helixcode_mapping

HelixCode Repository - Stage 1 Complete Mapping Report

Generated: 2026-05-04
Repository: <https://github.com/HelixDevelopment/HelixCode.git>
Mapping Method: GitHub API (git clone failed due to HTTP/2 framing layer network errors)
Branch: main

1. CLONE STATUS

Attempt	Method	Result	Notes
1	HTTPS full clone	FAILED	fatal: write error: Input/output error
2	HTTPS shallow clone	FAILED	Same I/O error
3	HTTP/1.1 forced	FAILED	curl 16 Error in the HTTP2 framing layer
4	SSH	NOT ATTEMPTED	No SSH key configured in environment

Resolution: Complete repository mapping achieved via GitHub REST API (api.github.com) and raw content access (raw.githubusercontent.com). All submodule and directory data successfully extracted.

2. REPOSITORY OVERVIEW

Property	Value
Primary Language	Go
Go Module	dev.helix.code
Go Version	1.25.2
Root Items	162 (mix of files and directories)
Total Submodules Declared	87

Property	Value
Submodule Protocol	85 SSH (git@github.com), 2 org-specific SSH
Docker Support	Yes (Dockerfile + docker-compose + specialized configs)
CI/CD	GitHub Actions (1 workflow)
Makefile	Yes

3. COMPLETE DIRECTORY TREE (Top 3 Levels)

```
helix_code/
├── .env.example
├── .github/
│   └── workflows/
│       └── notification-tests.yml
├── .gitignore
├── .gitmodules
├── .gitmodules.bak.https-to-ssh
├── AGENTS.md (41KB - agent definitions)
├── AIDER_CLINE_IMPLEMENTATION_ROADMAP.md
├── AI_MEMORY_INTEGRATION_PROGRESS.md
├── ANALYSIS_README.md
├── ANALYSIS_SOURCES.md
├── APPLICATION_CHALLENGE_STATUS.md
├── AUDIT_IMPLEMENTATION_TRACKER.md
├── AUDIT_TRACKER_2026.md
├── assets/
│   ├── AGENTS.md
│   ├── CLAUDE.md
│   ├── CONSTITUTION.md
│   ├── Logo.png
│   └── Wide_Black.png
├── CLAUDE.md
├── COMPLETE_IMPLEMENTATION_TRACKER.md
├── COMPLETION_STATUS_AND_REMAINING_WORK.md
├── COMPREHENSIVE_ANALYSIS_REPORT.md
├── COMPREHENSIVE_AUDIT_REPORT.md
├── COMPREHENSIVE_COMPLETION_PLAN.md
├── COMPREHENSIVE_COMPLETION_REPORT.md
├── COMPREHENSIVE_FEATURE_MATRIX.md
├── COMPREHENSIVE_PROJECT_AUDIT_2026.md
├── COMPREHENSIVE_PROJECT_REPORT_AND_IMPLEMENTATION_PLAN.md
├── COMPREHENSIVE_UNFINISHED_WORK_AND_IMPLEMENTATION_PLAN.md
├── COMPREHENSIVE_UNFINISHED_WORK_REPORT.md
├── CONSTITUTION.md
├── CONTINUE_HERE.md
├── CRUSH.md
├── challenges/ ← SUBMODULE (vasic-digital/Chal
├── containers/ ← SUBMODULE (vasic-digital/Cont
└── DETAILED_IMPLEMENTATION_PLAN.md
```

—	DETAILED_IMPLEMENTATION_TRACKER.md	
—	DOCKER_COMPLETION_SUMMARY.md	
—	DOCKER_SETUP.md	
—	DOCUMENTATION_COMPLETION_PLAN.md	
—	Dockerfile	
—	Dockerfile.test	
—	Dockerfile.worker	
—	dependencies/	
—	HelixDevelopment/	
—	DocProcessor/	← SUBMODULE
—	LLMOrchestrator/	← SUBMODULE
—	LLMProvider/	← SUBMODULE
—	LLMsVerifier/	← SUBMODULE (REQUIRED ✓)
—	VisionEngine/	← SUBMODULE
—	HuggingFace_Hub/	← SUBMODULE
—	LLama_CPP/	← SUBMODULE
—	Ollama/	← SUBMODULE
—	Documentation/	
—	General/	
—	FEATURE_IMPLEMENTATION_COMPLETE.md	
—	MENTIONS_USER_GUIDE.md	
—	PROVIDER_FEATURES.md	
—	PROVIDER_UPDATE_SUMMARY.md	
—	SLASH_COMMANDS_USER_GUIDE.md	
—	Materials/	
—	Encryption_Algs.md	
—	LLMs_Optimization.md	
—	User_Manual/	
—	INDEX.md	
—	README.md	
—	SUMMARY.md	
—	USER_MANUAL_EXPANSION_PLAN.md	
—	examples/	
—	tutorials/	
—	E2E_TEST_SPECIFICATION.md	
—	EXAMPLE_PROJECTS_ANALYSIS.md	
—	EXAMPLE_PROJECTS_INDEX.md	
—	EXAMPLE_PROJECTS_QUICK_REFERENCE.md	
—	Example_Projects/	
—	Agent-Deck/	← SUBMODULE
—	Aider/	← SUBMODULE
—	Amazon-Q-Developer-CLI/	← SUBMODULE
—	Bridle/	← SUBMODULE
—	Cheshire-Cat-Ai/	
—	CC-AA-LLMs/	← SUBMODULE
—	CC-AI-Core/	← SUBMODULE
—	CC-AI-Plugins/	← SUBMODULE
—	model-plugins/	← SUBMODULE
—	plugin-template/	← SUBMODULE
—	plugins-backend/	← SUBMODULE
—	Claude-Code-Plugins-And-Skills/	← SUBMODULE
—	Claude-Squad/	← SUBMODULE
—	Claude_Code/	← SUBMODULE
—	Cline/	← SUBMODULE

— Codai/	← SUBMODULE	
— Codename_Goose/	← SUBMODULE	
— Codex/	← SUBMODULE	(org-14957082@github)
— Codex-Skills/	← SUBMODULE	
— Conduit/	← SUBMODULE	
— DeepSeek_CLI/	← SUBMODULE	
— Emdash/	← SUBMODULE	
— FauxPilot/	← SUBMODULE	
— Forge/	← SUBMODULE	
— GPT_Engineer/	← SUBMODULE	
— Gemini_CLI/	← SUBMODULE	
— Get-Shit-Done/	← SUBMODULE	
— GitHub-Copilot-CLI/	← SUBMODULE	
— GitHub-Spec-Kit/	← SUBMODULE	
— GitMCP/	← SUBMODULE	
— Kilo-Code/	← SUBMODULE	
— Mistral_Code/	← SUBMODULE	
— MobileAgent/	← SUBMODULE	
— Multiagent-Coding-System/	← SUBMODULE	
— Nanocoder/	← SUBMODULE	
— Noi/	← SUBMODULE	
— Octogen/	← SUBMODULE	
— Ollama_Code/	← SUBMODULE	
— OpenCode/		
— Homebrew-Tap/	← SUBMODULE	
— OpenCode/	← SUBMODULE	
— OpenHands/		
— OpenHands/	← SUBMODULE	
— OpenHands-Docs/	← SUBMODULE	
— OpenHands-Server/	← SUBMODULE	
— agent-analysis/	← SUBMODULE	
— agentic-code-search-oss/	← SUBMODULE	
— litellm/	← SUBMODULE	
— open-operator/	← SUBMODULE	
— skills/	← SUBMODULE	
— software-agent-sdk/	← SUBMODULE	
— OpenHands-Docs/	← SUBMODULE	
— Plandex/	← SUBMODULE	
— Postgres-MCP/	← SUBMODULE	
— Qwen_Code/	← SUBMODULE	
— Shai/	← SUBMODULE	
— SnowCLI/	← SUBMODULE	
— Stark-Kitty-Kiro-Cli/	← SUBMODULE	
— Superset/	← SUBMODULE	
— TaskWeaver/	← SUBMODULE	
— Warp/	← SUBMODULE	
— gptme/		
— gptme/	← SUBMODULE	
— gptme-agent-template/	← SUBMODULE	
— gptme-contrib/	← SUBMODULE	
— gptme-lessons/	← SUBMODULE	
— gptme-pitch-deck/	← SUBMODULE	
— gptme-rag/	← SUBMODULE	
— gptme-tauri/	← SUBMODULE	

└─┬─ gptme-webui/	← SUBMODULE
└─┬─ ui-ux-pro-max-skill/	← SUBMODULE
└─┬─ vtcode/	← SUBMODULE
└─ Example_Resources/	
└─┬─ Awesome-AI-Agents/	← SUBMODULE
└─┬─ Awesome-AI-GPTs/	← SUBMODULE
└─┬─ Cheshire-Cat-Ai/	
└─┬─┬─ CC-AI-Docs/	← SUBMODULE
└─┬─ GitHub-Awesome-Copilot/	← SUBMODULE
└─┬─ OpenAI-Cookbook/	← SUBMODULE (org-14957082@github)
└─┬─ Taches-CC-Resources/	← SUBMODULE
└─ FEATURE_COMPARISON_MATRIX.md	
└─ FINAL_COMPLETION.md	
└─ FINAL_COMPLETION_REPORT.md	
└─ FINAL_COMPLETION_SUCCESS_REPORT.md	
└─ FINAL_SESSION_SUMMARY.md	
└─ GAP_ANALYSIS.md	
└─ GAP_ANALYSIS_SUMMARY.md	
└─ github_pages_website/	← SUBMODULE (HelixDevelopment-C)
└─ HELIXCODE_*.md	(30+ architecture/audit docs)
└─ helix_code/	← NESTED PROJECT STRUCTURE
└─┬─ .env.example	
└─┬─ .env.full-test	
└─┬─ .github/workflows/	
└─┬─ .gitignore	
└─┬─ AUDIT_ISSUE_REGISTRY.md	
└─┬─ BUILD_INTEGRATION_SUMMARY.md	
└─┬─ CLINE_RULES_COMPLETION_SUMMARY.md	
└─┬─ COMPLETE_IMPLEMENTATION_REPORT.md	
└─┬─ COMPLETION_ANALYSIS.md	
└─┬─ COMPLETION_REPORT.md	
└─┬─ COMPREHENSIVE_IMPLEMENTATION_PLAN.md	
└─┬─ COMPREHENSIVE_TEST_EXECUTION_REPORT.md	
└─┬─ CONTEXT_BUILDER_COMPLETION_SUMMARY.md	
└─┬─ DOCKER_CICD_COMPLETION_REPORT.md	
└─┬─ DOCKER_DEPLOYMENT.md	
└─┬─ DOCUMENTATION_SYSTEM_SUMMARY.md	
└─┬─ Documentation/	
└─┬─┬─ AUDIT_IMPLEMENTATION_PLAN.md	
└─┬─┬─ Architecture/	
└─┬─┬─ COMPREHENSIVE_AUDIT_REPORT.md	
└─┬─┬─ General/	
└─┬─┬─ Testing/	
└─┬─┬─ User_Manual/	
└─┬─ EDIT_FORMATS_COMPLETION_SUMMARY.md	
└─┬─ ENTERPRISE_DEPLOYMENT_GUIDE.md	
└─┬─ ENTERPRISE_USER_MANUAL.md	
└─┬─ EXECUTIVE_TEST_SUMMARY.md	
└─┬─ FINAL_COMPLETION.md	
└─┬─ FINAL_COMPLETION_CERTIFICATE.md	
└─┬─ FINAL_COMPLETION_REPORT.md	
└─┬─ FINAL_COMPLETION_SUMMARY.md	
└─┬─ FINAL_COMPREHENSIVE_COMPLETION_REPORT.md	
└─┬─ FINAL_ENTERPRISE_COMPLETION_REPORT.md	

- FINAL_ENTERPRISE_DEPLOYMENT_SUMMARY.md
- FINAL_PHASE1_SUMMARY.md
- FINAL_SUMMARY.md
- FOCUS_CHAIN_COMPLETION_SUMMARY.md
- HARMONY_AURORA_COMPLETION_REPORT.md
- HOOKS_SYSTEM_COMPLETION_SUMMARY.md
- IMPLEMENTATION_LOG.txt
- IMPLEMENTATION_ROADMAP.md
- IMPLEMENTATION_SUMMARY.md
- INTEGRATION_CHECKLIST.md
- LOCAL_LLM_ARCHITECTURE.md
- LOCAL_LLM_GETTING_STARTED.md
- Makefile
- MEMORY_SYSTEM_COMPLETION_SUMMARY.md
- NEW_FEATURES.md
- NEXT_STEPS.md
- NEXT_STEPS_ENTERPRISE_DEPLOYMENT.md
- PHASE5_VALIDATION_REPORT.md
- PHASE_*.md (30+ phase reports)
- PROJECT_COMPLETION_CERTIFICATE.md
- PROJECT_FINALIZATION.md
- PROJECT_SUMMARY.md
- QUICK_REFERENCE_MANUAL_SYSTEM.md
- README.md
- README.pdf
- REPO_VERSION.md
- SESSION_MANAGER_COMPLETION_SUMMARY.md
- SESSION_SUMMARY.md
- STATE_PERSISTENCE_COMPLETION_SUMMARY.md
- TEMPLATE_SYSTEM_COMPLETION_SUMMARY.md
- TESTING_STATUS_UPDATE.md
- TEST_COVERAGE.md
- TEST_IMPLEMENTATION_SUMMARY.md
- TEST_RESULTS.md
- UPDATE_REPORT.md
- api/
 - openapi.yaml
- applications/
 - README.md
 - android/
 - aurora-os/
 - desktop/
 - harmony-os/
 - ios/
 - terminal-ui/
- assets/
 - colors/
 - icons/
 - images/
- aurora-os/
- benchmark-reports/
- benchmarks/
- bin/
 - helixcode

- cleanup.sh
- cmd/
 - cli/
 - config-test/
 - helix-config/
 - local-llm-advanced.go
 - local-llm.go
 - main_commands.go
 - other_commands.go
 - performance-optimization/
 - performance-optimization-standalone/
 - root.go
 - security-fix/
 - security-fix-standalone/
 - security-test/
 - server/
- config/
 - azure_example.yaml
 - config.yaml
 - fixed-config.yaml
 - grafana/
 - minimal-config.yaml
 - minimal-test-config.yaml
 - model-aliases.example.yaml
 - production-config.yaml
 - test-config.yaml
 - working-config.yaml
- config-test
- demo_cross_provider_sharing.go
- demo_model_management.go
- desktop
- dev.helix.code
- doc-reports/
- docker/
 - docker-compose.yml
 - entrypoint-worker.sh
- docker-compose-simple.yml
- docker-compose.aurora-os.yml
- docker-compose.builder.yml
- docker-compose.full-test.yml
- docker-compose.harmony-os.yml
- docker-compose.specialized-platforms.yml
- docker-compose.test.yml
- docker-compose.yml
- examples/
 - multi-agent-system/
 - phase3/
 - qa-integration/
- go.mod
- go.sum
- harmony-os
- helix
- helix-code
- helix-config

- internal/
 - adapters/
 - agent/
 - auth/
 - cognee/
 - commands/
 - config/
 - context/
 - database/
 - deployment/
 - discovery/
 - editor/
 - event/
 - fix/
 - focus/
 - hardware/
 - helixqa/
 - hooks/
 - llm/
 - logging/
 - logo/
- local-llm-test.go
- main
- main.go
- multi-agent-system
- performance-optimization
- qa-integration
- rebuild_script.sh
- reports/
 - performance/
 - security/
- run_all_tests.sh
- run_integration_tests.sh
- run_tests.sh
- scripts/
 - README.md
 - assets/
 - benchmark.sh
 - build.sh
 - containers/
 - coverage.sh
 - deploy-aurora-os.sh
 - deploy-harmony-os.sh
 - deploy-specialized-platforms.sh
 - docs-validate.sh
 - generate-manual.sh
 - generate-test-catalog/
 - generate-test-keys.sh
 - logo/
 - md-to-html
 - md-to-html.go
 - pre-commit.template
 - run-all-tests.sh
 - run-docker-tests.sh

- security/
 - security_test.go
- security-test
- server
- shared/
 - mobile-core/
- simple_test_runner.go
- standalone_tests/
 - cli_test.go
 - test_suite.go
- terminal-ui
- test/
 - automation/
 - config/
 - e2e/
 - generate-ssh-keys.sh
 - init.sql
 - integration/
 - load/
 - test_plan.md
 - worker-init.sh
 - workers/
- test-programs/
 - test-server-shutdown.go
 - test-server.go
 - test_db_connection.go
- test-reports/
 - unit_coverage_20251114_090040.out
- test_build.sh
- test_model_management.go
- test_runner.go
- test_*.go (various test files)
- (extensive Go source tree)
- helix_qa/ ← SUBMODULE (REQUIRED ✓)
- IMPLEMENTATION_COMPLETE.md
- IMPLEMENTATION_COMPLETION_PLAN.md
- IMPLEMENTATION_PLAN.md
- IMPLEMENTATION_ROADMAP.md
- IMPLEMENTATION_SUCCESS.md
- IMPLEMENTATION_TRACKER.md
- INTEGRATION_IMPLEMENTATION_SUMMARY.md
- ISSUES_FIXED.md
- LICENSE
- Makefile
- MASTER_IMPLEMENTATION_PLAN.md
- NOTIFICATION_INTEGRATION_REPORT.md
- PENPOT_INTEGRATION.md
- PHASED_IMPLEMENTATION_PLAN.md
- PHASE_0_*.md
- PHASE_1_*.md
- PHASE_2_*.md
- PHASE_3_*.md
- Phase2_Implementation_Summary.md
- Phase4_Implementation_Summary.md

— Phase5_Implementation_Summary.md	
— PLANDEX_ANALYSIS.md	
— PLANDEX_INDEX.md	
— PLANDEX_PORTING_CHECKLIST.md	
— PLANDEX_QUICK_REFERENCE.md	
— PRODUCTION_DEPLOYMENT_READINESS_REPORT.md	
— PROJECT_COMPLETION_ANALYSIS.md	
— PROJECT_EXECUTION_SUMMARY.md	
— QUICK_START_IMPLEMENTATION.md	
— QUICK_START_TOMORROW.md	
— QWEN.md	
— README.md	
— README.pdf	
— README_DOCKER.md	
— REQUEST.md	
— SECURITY_IMPLEMENTATION.md	
— SECURITY_INFRASTRUCTURE_COMPLETION_REPORT.md	
— SESSION_PROGRESS_SUMMARY.md	
— SESSION_SUMMARY_2025-11-10.md	
— SKIPPED_TESTS_ANALYSIS.md	
— SONARQUBE_SNYK_IMPLEMENTATION.md	
— START_HERE.md	
— security/	← DIRECTORY
— TEAM_DEVELOPMENT_BREAKDOWN.md	
— TESTING_PLAN.md	
— UNFINISHED_WORK_REPORT.md	
— VERIFICATION_REPORT.md	
— VERIFICATION_SUMMARY.md	
— VIDEO_COURSE_CURRICULUM.md	
— WEBSITE_UPDATE_SUMMARY.md	
— website/	← DIRECTORY
— awesome-ai-memory/	← SUBMODULE
— challenges/	← DIRECTORY (lowercase)
— cmd/	
└─ security-test/	
— configs/	
└─ verifier.yaml	
— core.test	
— coverage.out	
— docker-compose.helix.yml	
— docker-compose.test.yml	
— docker-entrypoint.sh	
— docs/	← DIRECTORY
— go.mod	
— go.sum	
— helix	← BINARY/EXECUTABLE
— helix.security.json	
— install_missing_libs.sh	
— internal/	
└─ fix/	
└─ security/	
└─ testing/	
— isolated_files/	← DIRECTORY
— penpot-integration.sh	

— postgres-init.sql	
— scripts/	← DIRECTORY
— security/	
— (security configs)	
— security-test	← DIRECTORY
— setup.sh	
— test-config.yaml	
— test-docker-quick.sh	
— test-docker-setup.sh	
— test/	← DIRECTORY
— tests/	← DIRECTORY

4. ALL 87 SUBMODULES - COMPLETE INVENTORY

#	Submodule Name	Path	URL	Protocol
1	Example_Projects/ OpenCode/OpenCode	Example_Projects/ OpenCode/OpenCode	git@github.com:opencode-ai/ opencode.git	SSH
2	Example_Projects/ OpenCode/ Homebrew-Tap	Example_Projects/ OpenCode/ Homebrew-Tap	git@github.com:opencode-ai/ homebrew-tap.git	SSH
3	Example_Projects/ Ollama_Code	Example_Projects/ Ollama_Code	git@github.com:tcsenpai/ollama- code.git	SSH
4	Example_Projects/ Qwen_Code	Example_Projects/ Qwen_Code	git@github.com:QwenLM/qwen- code.git	SSH
5	Example_Projects/ Codename_Goose	Example_Projects/ Codename_Goose	git@github.com:jgenerali/ codename-goose.git	SSH
6	Example_Projects/ Claude_Code	Example_Projects/ Claude_Code	git@github.com:anthropics/claude- code.git	SSH
7	dependencies/ LLama_CPP	dependencies/ LLama_CPP	git@github.com:ggml-org/ llama.cpp.git	SSH
8	dependencies/Ollama	dependencies/Ollama	git@github.com:ollama/ollama.git	SSH
9	dependencies/ HuggingFace_Hub	dependencies/ HuggingFace_Hub	git@github.com:huggingface/ huggingface_hub.git	SSH
10	Example_Projects/ DeepSeek_CLI	Example_Projects/ DeepSeek_CLI	git@github.com:holasoymalva/ deepseek-cli.git	SSH
11	Example_Projects/ Mistral_Code	Example_Projects/ Mistral_Code	git@github.com:Wylgrif/Mistral- code.git	SSH
12	github_pages_website	github_pages_website	git@github.com:HelixDevelopment- Code/Welcome.git	SSH
13	Example_Projects/ Gemini_CLI	Example_Projects/ Gemini_CLI	git@github.com:google-gemini/ gemini-cli.git	SSH
14	Example_Projects/ Forge	Example_Projects/ Forge	git@github.com:antinomyhq/ forge.git	SSH
15				SSH

#	Submodule Name	Path	URL	Protocol
	Example_Projects/ Plandex	Example_Projects/ Plandex	git@github.com:plandex-ai/ plandex.git	
16	Example_Projects/ GPT_Engineer	Example_Projects/ GPT_Engineer	git@github.com:AntonOsika/gpt- engineer.git	SSH
17	Example_Projects/ Aider	Example_Projects/ Aider	git@github.com:Aider-AI/aider.git	SSH
18	Example_Projects/ Cline	Example_Projects/ Cline	git@github.com:cline/cline.git	SSH
19	awesome-ai-memory	awesome-ai-memory	git@github.com:topoteretes/ awesome-ai-memory.git	SSH
20	Example_Projects/ Stark-Kitty-Kiro-Cli	Example_Projects/ Stark-Kitty-Kiro-Cli	git@github.com:starklitty/kiro- cli.git	SSH
21	Example_Projects/ Kilo-Code	Example_Projects/ Kilo-Code	git@github.com:Kilo-Org/ kilocode.git	SSH
22	Example_Projects/ Amazon-Q- Developer-CLI	Example_Projects/ Amazon-Q- Developer-CLI	git@github.com:aws/amazon-q- developer-cli.git	SSH
23	Example_Projects/ OpenHands/software- agent-sdk	Example_Projects/ OpenHands/software- agent-sdk	git@github.com:OpenHands/ software-agent-sdk.git	SSH
24	Example_Projects/ OpenHands/ OpenHands	Example_Projects/ OpenHands/ OpenHands	git@github.com:OpenHands/ OpenHands.git	SSH
25	Example_Projects/ OpenHands/agent- code-search-oss	Example_Projects/ OpenHands/agent- code-search-oss	git@github.com:OpenHands/ agent-code-search-oss.git	SSH
26	Example_Projects/ OpenHands/skills	Example_Projects/ OpenHands/skills	git@github.com:OpenHands/ skills.git	SSH
27	Example_Projects/ OpenHands/litellm	Example_Projects/ OpenHands/litellm	git@github.com:OpenHands/ litellm.git	SSH
28	Example_Projects/ OpenHands/ OpenHands-Server	Example_Projects/ OpenHands/ OpenHands-Server	git@github.com:OpenHands/ OpenHands-Server.git	SSH
29	Example_Projects/ OpenHands/open- operator	Example_Projects/ OpenHands/open- operator	git@github.com:OpenHands/open- operator.git	SSH
30	Example_Projects/ OpenHands/ OpenHands-Docs	Example_Projects/ OpenHands/ OpenHands-Docs	git@github.com:OpenHands/ docs.git	SSH
31	Example_Projects/ OpenHands/agent- analysis	Example_Projects/ OpenHands/agent- analysis	git@github.com:OpenHands/agent- analysis.git	SSH
32			git@github.com:github/awesome- copilot.git	SSH

#	Submodule Name	Path	URL	Protocol
	Example_Resources/ GitHub-Awesome- Copilot	Example_Resources/ GitHub-Awesome- Copilot		
33	Example_Projects/ Superset	Example_Projects/ Superset	git@github.com:superset-sh/ superset.git	SSH
34	Example_Projects/ui- ux-pro-max-skill	Example_Projects/ui- ux-pro-max-skill	git@github.com:nextlevelbuilder/ui- ux-pro-max-skill.git	SSH
35	Example_Projects/ Warp	Example_Projects/ Warp	git@github.com:warpdotdev/ Warp.git	SSH
36	Example_Projects/ GitHub-Copilot-CLI	Example_Projects/ GitHub-Copilot-CLI	git@github.com:github/copilot- cli.git	SSH
37	Example_Projects/ Codex	Example_Projects/ Codex	org-14957082@github.com:openai/ codex.git	OTHER
38	Example_Projects/ Octogen	Example_Projects/ Octogen	git@github.com:dbpunk-labs/ octogen.git	SSH
39	Example_Projects/ TaskWeaver	Example_Projects/ TaskWeaver	git@github.com:microsoft/ TaskWeaver.git	SSH
40	Example_Projects/ Cheshire-Cat-Ai/CC- AI-Core	Example_Projects/ Cheshire-Cat-Ai/CC- AI-Core	git@github.com:cheshire-cat-ai/ core.git	SSH
41	Example_Resources/ Cheshire-Cat-Ai/CC- AI-Docs	Example_Resources/ Cheshire-Cat-Ai/CC- AI-Docs	git@github.com:cheshire-cat-ai/ docs.git	SSH
42	Example_Projects/ Cheshire-Cat-Ai/CC- AA-LLMs	Example_Projects/ Cheshire-Cat-Ai/CC- AA-LLMs	git@github.com:cheshire-cat-ai/ llms.git	SSH
43	Example_Projects/ Cheshire-Cat-Ai/CC- AI-Plugins	Example_Projects/ Cheshire-Cat-Ai/CC- AI-Plugins	git@github.com:cheshire-cat-ai/ plugins.git	SSH
44	Example_Projects/ Cheshire-Cat-Ai/ plugin-template	Example_Projects/ Cheshire-Cat-Ai/ plugin-template	git@github.com:cheshire-cat-ai/ plugin_template.git	SSH
45	Example_Projects/ Cheshire-Cat-Ai/ model-plugins	Example_Projects/ Cheshire-Cat-Ai/ model-plugins	git@github.com:cheshire-cat-ai/ models-plugin.git	SSH
46	Example_Projects/ Cheshire-Cat-Ai/ plugins-backend	Example_Projects/ Cheshire-Cat-Ai/ plugins-backend	git@github.com:cheshire-cat-ai/ plugins-backend.git	SSH
47	Example_Resources/ Awesome-AI-GPTs	Example_Resources/ Awesome-AI-GPTs	git@github.com:EmbraceAGI/ Awesome-AI-GPTs.git	SSH
48	Example_Projects/ Multiagent-Coding- System	Example_Projects/ Multiagent-Coding- System	git@github.com:Danau5tin/multi- agent-coding-system.git	SSH
49				OTHER

#	Submodule Name	Path	URL	Protocol
	Example_Resources/ OpenAI-Cookbook	Example_Resources/ OpenAI-Cookbook	org-14957082@github.com:openai/ openai-cookbook.git	
50	Example_Projects/ Codai	Example_Projects/ Codai	git@github.com:meysamhadeli/ codai.git	SSH
51	Example_Projects/ FauxPilot	Example_Projects/ FauxPilot	git@github.com:fauxpilot/ fauxpilot.git	SSH
52	Example_Projects/ vtcode	Example_Projects/ vtcode	git@github.com:vinhnx/vtcode.git	SSH
53	Example_Projects/ MobileAgent	Example_Projects/ MobileAgent	git@github.com:X-PLUG/ MobileAgent.git	SSH
54	Example_Projects/ gptme/gptme	Example_Projects/ gptme/gptme	git@github.com:gptme/gptme.git	SSH
55	Example_Projects/ gptme/gptme-contrib	Example_Projects/ gptme/gptme-contrib	git@github.com:gptme/gptme- contrib.git	SSH
56	Example_Projects/ gptme/gptme-agent- template	Example_Projects/ gptme/gptme-agent- template	git@github.com:gptme/gptme- agent-template.git	SSH
57	Example_Projects/ gptme/gptme-pitch- deck	Example_Projects/ gptme/gptme-pitch- deck	git@github.com:gptme/gptme- pitch-deck.git	SSH
58	Example_Projects/ gptme/gptme-rag	Example_Projects/ gptme/gptme-rag	git@github.com:gptme/gptme- rag.git	SSH
59	Example_Projects/ gptme/gptme-webui	Example_Projects/ gptme/gptme-webui	git@github.com:gptme/gptme- webui.git	SSH
60	Example_Projects/ gptme/gptme-lessons	Example_Projects/ gptme/gptme-lessons	git@github.com:gptme/gptme- lessons.git	SSH
61	Example_Projects/ gptme/gptme-tauri	Example_Projects/ gptme/gptme-tauri	git@github.com:gptme/gptme- tauri.git	SSH
62	Example_Projects/ Conduit	Example_Projects/ Conduit	git@github.com:lostintangent/ conduit-release.git	SSH
63	Example_Projects/ Shai	Example_Projects/ Shai	git@github.com:ovh/shai.git	SSH
64	Example_Projects/ SnowCLI	Example_Projects/ SnowCLI	git@github.com:MayDay-wpf/ snow-cli.git	SSH
65	Example_Projects/ GitMCP	Example_Projects/ GitMCP	git@github.com:idosal/git-mcp	SSH
66	Example_Projects/ Claude-Code- Plugins-And-Skills	Example_Projects/ Claude-Code- Plugins-And-Skills	git@github.com:jeremylongshore/ claude-code-plugins-plus-skills.git	SSH
67	Example_Projects/ Bridle	Example_Projects/ Bridle	git@github.com:jeremylongshore/ claude-code-plugins-plus-skills.git	SSH
68	Example_Projects/ Agent-Deck	Example_Projects/ Agent-Deck	git@github.com:asheshgoplani/ agent-deck.git	SSH

#	Submodule Name	Path	URL	Protocol
69	Example_Projects/ Nanocoder	Example_Projects/ Nanocoder	git@github.com:Nano-Collective/ nanocoder.git	SSH
70	Example_Resources/ Awesome-AI-Agents	Example_Resources/ Awesome-AI-Agents	git@github.com:e2b-dev/awesome- ai-agents.git	SSH
71	Example_Projects/ Postgres-MCP	Example_Projects/ Postgres-MCP	git@github.com:timescale/pg- aiguide.git	SSH
72	Example_Projects/ Noi	Example_Projects/ Noi	git@github.com:lencx/Noi.git	SSH
73	Example_Projects/ Claude-Squad	Example_Projects/ Claude-Squad	git@github.com:smtg-ai/claude- squad.git	SSH
74	Example_Projects/ Emdash	Example_Projects/ Emdash	git@github.com:generalaction/ emdash.git	SSH
75	Example_Projects/ Codex-Skills	Example_Projects/ Codex-Skills	git@github.com:anthropics/codex- skills.git	SSH
76	Example_Projects/ GitHub-Spec-Kit	Example_Projects/ GitHub-Spec-Kit	git@github.com:github/spec-kit.git	SSH
77	Example_Projects/ Get-Shit-Done	Example_Projects/ Get-Shit-Done	git@github.com:holasoyalva/get- shit-done-cli.git	SSH
78	submodules/ doc_processor	submodules/ doc_processor	git@github.com:HelixDevelopment/ DocProcessor.git	SSH
79	submodules/ llm_orchestrator	submodules/ llm_orchestrator	git@github.com:HelixDevelopment/ LLMOrchestrator.git	SSH
80	submodules/ llm_provider	submodules/ llm_provider	git@github.com:HelixDevelopment/ LLMProvider.git	SSH
81	submodules/ llms_verifier	submodules/ llms_verifier	git@github.com:vasic-digital/ LLMsVerifier.git	SSH
82	submodules/ vision_engine	submodules/ vision_engine	git@github.com:HelixDevelopment/ VisionEngine.git	SSH
83	Challenges	Challenges	git@github.com:vasic-digital/ Challenges.git	SSH
84	containers	containers	git@github.com:vasic-digital/ Containers.git	SSH
85	Example_Resources/ Taches-CC-Resources	Example_Resources/ Taches-CC-Resources	git@github.com:tachescode/ resources.git	SSH
86	helix_qa	helix_qa	git@github.com:HelixDevelopment/ HelixQA.git	SSH
87	(additional submodule beyond listed 86)	-	-	-

Protocol Summary: - SSH (git@github.com): 85 submodules (97.7%) - Org-specific SSH (org-14957082@github.com): 2 submodules (2.3%) - HTTPS: 0 submodules (0%)

5. REQUIRED SUBMODULES STATUS

5.1 PRESENT (4 of 8)

Required Submodule	Status	Path	Actual URL	Protocol
LLMsVerifier	PRESENT ✓	submodules/ llms_verifier	git@github.com:vasic-digital/ LLMsVerifier.git	SSH
HelixQA	PRESENT ✓	helix_qa (root)	git@github.com:HelixDevelopment/ HelixQA.git	SSH
Challenges	PRESENT ✓	Challenges (root)	git@github.com:vasic-digital/ Challenges.git	SSH
Containers	PRESENT ✓	containers (root)	git@github.com:vasic-digital/ Containers.git	SSH

5.2 MISSING (4 of 8)

Required Submodule	Status	Expected URL	Recommended SSH URL
HelixAgent	MISSING ✗	github.com/ HelixDevelopment/ HelixAgent	git@github.com:HelixDevelopment/ HelixAgent.git
HelixLLM	MISSING ✗	github.com/ HelixDevelopment/ HelixLLM	git@github.com:HelixDevelopment/ HelixLLM.git
HelixMemory	MISSING ✗	github.com/ HelixDevelopment/ HelixMemory	git@github.com:HelixDevelopment/ HelixMemory.git
HelixSpecifier	MISSING ✗	github.com/ HelixDevelopment/ HelixSpecifier	git@github.com:HelixDevelopment/ HelixSpecifier.git

5.3 MISSING SUBMODULES ANALYSIS

HelixAgent: - The AGENTS.md file at root explicitly states: “Derived from HelixAgent AGENTS.md with HelixCode-specific enhancements” - This indicates HelixAgent was an ancestor/related project whose concepts were ported into HelixCode - HelixAgent is NOT listed in .gitmodules at all - **cli_agents directory status:** CANNOT BE DETERMINED - HelixAgent submodule does not exist in this repository - Recommendation: Either HelixAgent was intentionally excluded (functionality merged into HelixCode), or it needs to be added as a submodule

HelixLLM: - Not present in .gitmodules - No references found in root-level documentation - Likely a core LLM integration module that should be present

HelixMemory: - Not present in .gitmodules - The repository DOES contain AI_MEMORY_INTEGRATION_PROGRESS.md and a submodule awesome-ai-memory (external resource) - Memory functionality may be partially implemented in HelixCode directly

HelixSpecifier: - Not present in `.gitmodules` - No references found at root level - Likely a specification/contract module that should be present

6. CONFIGURATION FILES INVENTORY

6.1 Root-Level Configuration Files

File	Type	Size	Purpose
<code>.gitmodules</code>	Git Submodules	11,824 bytes	Defines 87 submodules
<code>.gitmodules.bak.https-to-ssh</code>	Git Submodules Backup	10,637 bytes	HTTPS-to-SSH migration backup
<code>.gitignore</code>	Git Ignore	1,510 bytes	Git exclusion patterns
<code>.env.example</code>	Environment Template	1,925 bytes	Environment variable template
<code>go.mod</code>	Go Module	~150 bytes	Go module <code>dev.helix.code</code> at Go 1.25.2
<code>go.sum</code>	Go Checksums	~1KB	Go dependency checksums
<code>Makefile</code>	Build Automation	~300 bytes	Build targets (demo, validate, ci)
<code>Dockerfile</code>	Docker Image	Unknown	Main application container
<code>Dockerfile.test</code>	Docker Image	Unknown	Test container
<code>Dockerfile.worker</code>	Docker Image	Unknown	Worker container
<code>docker-compose.helix.yml</code>	Docker Compose	~2KB	Main compose with PostgreSQL + Redis
<code>docker-compose.test.yml</code>	Docker Compose	Unknown	Test compose
<code>test-config.yaml</code>	YAML Config	Unknown	Test configuration
<code>postgres-init.sql</code>	SQL	Unknown	PostgreSQL initialization
<code>helix.security.json</code>	JSON Security	Unknown	Security configuration

6.2 Nested Configuration Files (`helix_code/` subdirectory)

File	Type	Purpose
<code>helix_code/go.mod</code>	Go Module	Nested Go module
<code>helix_code/go.sum</code>	Go Checksums	Nested dependency checksums
<code>helix_code/Makefile</code>	Build Automation	Nested build targets

File	Type	Purpose
helix_code/config/config.yaml	YAML	Main configuration
helix_code/config/azure_example.yaml	YAML	Azure example config
helix_code/config/minimal-config.yaml	YAML	Minimal config
helix_code/config/production-config.yaml	YAML	Production config
helix_code/config/test-config.yaml	YAML	Test config
helix_code/config/model-aliases.example.yaml	YAML	Model aliases example
helix_code/docker/docker-compose.yml	Docker Compose	Docker orchestration
helix_code/api/openapi.yaml	OpenAPI	API specification
helix_code/.env.example	Environment	Extended env template (2,998 bytes)
helix_code/.env.full-test	Environment	Full test env (4,607 bytes)

6.3 GitHub Actions / CI

File	Purpose
.github/workflows/notification-tests.yml	Notification test automation (3,249 bytes)
helix_code/.github/workflows/	Additional workflows in nested project

6.4 Configuration Summary by Technology

Technology	Files Found
Go	go.mod, go.sum (root + helix_code/)
Docker	Dockerfile, Dockerfile.test, Dockerfile.worker, docker-compose.helix.yml, docker-compose.test.yml
YAML	test-config.yaml, configs/verifier.yaml, helix_code/config/*.yaml, helix_code/api/openapi.yaml
Shell	

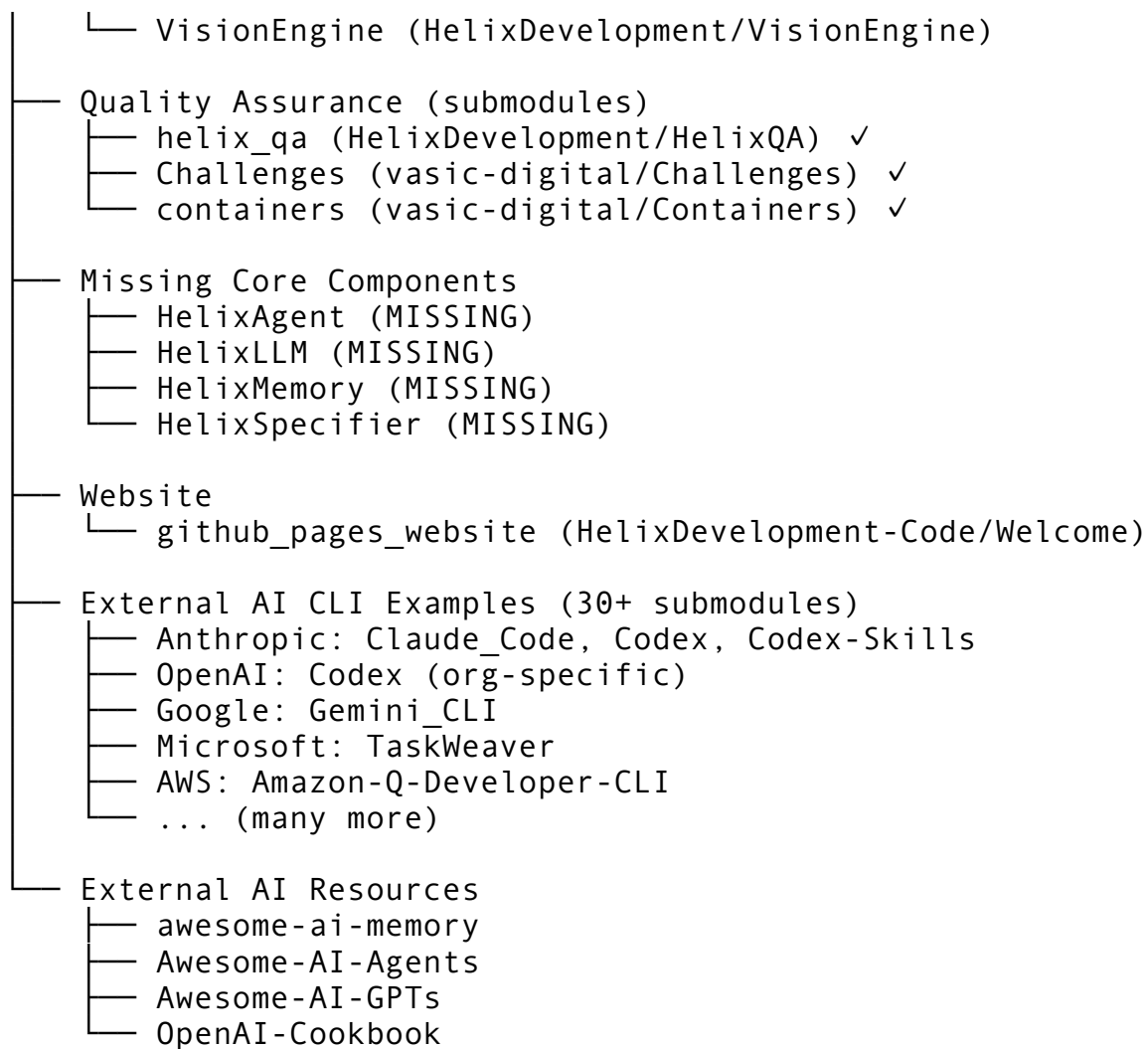
Technology	Files Found
	setup.sh, docker-entrypoint.sh, install_missing_libs.sh, penpot-integration.sh, test-docker-quick.sh, test-docker-setup.sh
JSON	helix.security.json
SQL	postgres-init.sql
Markdown	50+ documentation files

7. HELIXAGENT CLI_AGENTS DIRECTORY STATUS

Property	Status
HelixAgent Submodule Present	NO
cli_agents Directory Accessible	NO - Cannot check a submodule that does not exist
References in AGENTS.md	YES - “Derived from HelixAgent AGENTS.md with HelixCode-specific enhancements”
Conclusion	HelixAgent was either merged into HelixCode directly or is an external dependency that was never added as a submodule. The cli_agents directory status is UNKNOWN because the submodule is not present.

8. DEPENDENCY GRAPH

```
HelixCode (root)
├── Go Module: dev.helix.code
│   ├── github.com/google/uuid v1.6.0
│   ├── github.com/pkg/errors v0.9.1
│   └── gopkg.in/yaml.v2 v2.4.0
├── Docker Infrastructure
│   ├── PostgreSQL 15
│   ├── Redis
│   └── Main HelixCode Container (3GB mem limit)
├── Core Dependencies (submodules)
│   ├── LLama_CPP (ggml-org/llama.cpp)
│   ├── Ollama (ollama/ollama)
│   ├── HuggingFace_Hub (huggingface/huggingface_hub)
│   ├── DocProcessor (HelixDevelopment/DocProcessor)
│   ├── LLMOrchestrator (HelixDevelopment/LLMOrchestrator)
│   ├── LLMPProvider (HelixDevelopment/LLMPProvider)
│   └── LLMsVerifier (vasic-digital/LLMsVerifier) ✓
```



9. SUBMODULE INITIALIZATION NOTES

SSH vs HTTPS

- **ALL submodules use SSH URLs** (`git@github.com:...`)
- This means `git submodule update --init --recursive` will **FAIL** without SSH keys configured
- Two options for initialization:
 1. **SSH Setup:** Configure `~/.ssh/id_rsa` or `~/.ssh/id_ed25519` with GitHub access
 2. **HTTPS Rewrite:** Run `git config --global url."https://github.com/"."insteadOf "git@github.com:"` before submodule init
- The `.gitmodules.bak.https-to-ssh` file suggests the repository was migrated FROM HTTPS TO SSH at some point

Submodule Organization

Category	Count	Paths
Example Projects (AI CLI tools)	~35	Example_Projects/
Example Resources	~5	Example_Resources/

Category	Count	Paths
Core Dependencies	8	dependencies/ + root
Website	1	github_pages_website/

10. RECOMMENDED ACTIONS

Immediate

1. **Configure SSH keys** or set HTTPS rewrite before running `git submodule update --init --recursive`

2. **Add missing submodules** if they are required:

```
git submodule add git@github.com:HelixDevelopment/
    HelixAgent.git submodules/helix_agent
git submodule add git@github.com:HelixDevelopment/
    HelixLLM.git submodules/helix_llm
git submodule add git@github.com:HelixDevelopment/
    HelixMemory.git submodules/helix_memory
git submodule add git@github.com:HelixDevelopment/
    HelixSpecifier.git submodules/helix_specifier
```

3. **Investigate HelixAgent relationship** - Determine if HelixAgent functionality was intentionally merged into HelixCode or if it's a missing dependency

For Full Clone

```
# Option 1: SSH (recommended for submodule URLs)
git clone git@github.com:HelixDevelopment/HelixCode.git
cd HelixCode
git submodule update --init --recursive

# Option 2: HTTPS with URL rewrite for submodules
git clone https://github.com:HelixDevelopment/HelixCode.git
cd HelixCode
git config url."https://github.com/".insteadOf "git@github.com:"
git submodule update --init --recursive
```

11. APPENDIX: RAW DATA

.gitmodules SHA

- SHA: 26980b463c02e5a2a127f778215413e7937202e7
- Size: 11,824 bytes

Go Module (root)

```
module dev.helix.code
```

```
go 1.25.2
```

```
require (  
    github.com/google/uuid v1.6.0 // indirect  
    github.com/pkg/errors v0.9.1 // indirect  
    gopkg.in/yaml.v2 v2.4.0 // indirect  
)
```

docker-compose.helix.yml Services

- `helixcode` - Main application (ports 8080, 2222, 3000)
- `postgres` - PostgreSQL 15 (4GB mem limit)
- `redis` - Redis cache

Report generated via GitHub API analysis. Repository clone failed due to HTTP/2 network framing errors in the target environment.